

NutriVerse — 3 Visualization Proposal

Information Visualization — Food & Nutrition

Project: Information Visualization — Food & Nutrition

Concept: Three interconnected visualizations that explore nutrition from micro (molecules) → personal (your plate) → macro (the world)

The Narrative Arc

The three visualizations follow a **zoom-out storytelling structure**:

MICRO	PERSONAL	MACRO
Viz 1 "The Nutrient Cosmos"	Viz 2 "The Perfect Plate"	Viz 3 "What the World Is Missing"
How nutrients interact inside your body	What happens when you combine foods into a meal	Where are people deficient and what local foods could help
NETWORK GRAPH + detail panel	INTERACTIVE BUILDER + live radar analysis	CHOROPLETH MAP + linked bar charts

This structure demonstrates **range** (3 completely different chart types), **depth** (scientific research in Viz 1), and **storytelling** (micro → macro narrative).

Visualization 1: "The Nutrient Cosmos"

Core Question

"How do essential nutrients enhance or inhibit each other at dietary levels, and what does this mean for everyday food combinations?"

Sub-questions

1. Which nutrients have the most interactions (most "connected" in the network)?
2. Are there more synergies or antagonisms in a typical diet?
3. Which common meals unknowingly contain antagonistic nutrient pairs?
4. Can you trace a "chain reaction" — e.g., Vitamin D → Calcium → Iron (D helps Ca, but Ca blocks Fe)?

Description

An interactive **force-directed network graph** where:

- **Nodes** = essential nutrients (vitamins, minerals, macros), sized by number of connections
- **Edges** = scientifically documented interactions at food-level doses
- **Green edges** = synergies (enhances absorption/function)
- **Red dashed edges** = antagonisms (inhibits absorption)
- **Edge thickness** = strength of the interaction

Clicking a node or edge opens a **detail panel** showing:

- The biochemical mechanism in plain language
- The scientific source (journal citation)
- Real food combinations where this interaction occurs
- A mini radar chart of a food rich in the selected nutrients

Three layout modes (inspired by Sustainability Cosmos):

- **Network** — organic force-directed layout showing natural clustering
- **Radial** — nutrients arranged in a circle, edges crossing the center
- **Grouped** — clustered by category (Vitamins / Minerals / Macros)

Scope: Food-Level Interactions

This visualization focuses exclusively on interactions that are **scientifically documented at normal dietary doses** — i.e., quantities you would realistically consume in a meal, not mega-dose supplements.

Examples of included interactions (occur at food-level doses):

- **Vitamin C → Iron (synergy)**: Ascorbic acid increases non-heme iron absorption 2–6× — e.g., squeezing lemon on spinach.
- **Calcium → Iron (antagonism)**: Ca competes with Fe at the gut transporter — e.g., drinking milk with a steak.
- **Vitamin D → Calcium (synergy)**: Vit D upregulates calcium-binding proteins in the intestine.
- **Phytates → Zinc/Iron (antagonism)**: Phytic acid in whole grains chelates minerals, reducing uptake.
- **Fat → Vitamins A/D/E/K (synergy)**: Fat-soluble vitamins require dietary fat for absorption — e.g., eating carrots with olive oil.
- **Vitamin C → Vitamin E (synergy)**: Vit C regenerates oxidized Vit E — e.g., eating fruit with nuts.
- **Caffeine → Calcium (antagonism)**: Caffeine increases urinary calcium excretion.

Excluded (only relevant at supplement mega-doses):

- Zinc → Copper antagonism (requires >50mg Zn/day, impossible from food alone)
- Vitamin A toxicity interactions (only at retinol supplement doses, not dietary β -carotene)

- Iron → Zinc antagonism (mostly documented at supplemental iron >25mg)

This scoping still yields ~20–30 well-documented edges — more than sufficient for a rich, scientifically grounded network graph.

What Makes It Creative

- **Chain reaction highlighting:** Click a nutrient and see a ripple effect through connected nutrients → shows cascading impacts
- **”Meal X-Ray” mode:** Enter a meal (e.g., ”spinach salad with lemon dressing and milk”) and the network highlights all active interactions in that meal simultaneously — green and red edges light up
- Nobody has built a beautiful, interactive nutrient interaction network before — this is the **standout feature**

Data Sources

Source	What it provides
Deanna Minich Interactions Chart	Matrix of all vitamin-mineral interactions
Linus Pauling Institute	Evidence-based interaction details per nutrient
NIH ODS Fact Sheets	Documented interactions + DRI values
Manual curation	~20-30 food-level interactions with mechanisms + citations

Chart Type: Force-directed network graph + detail panel + radar chart

Visualization 2: ”The Perfect Plate”

Core Question

”If you could see the nutritional fingerprint of every food you eat, what would the ’perfect’ balanced meal actually look like — and how close are common meals?”

Sub-questions

1. What is the unique nutritional ”shape” (fingerprint) of individual foods?
2. How does adding or removing a food from a meal change its overall nutritional profile?
3. Which single foods cover the most nutritional ground (highest ”coverage score”)?
4. What’s the gap between a common fast-food meal vs. a balanced home-cooked meal?
5. Which food would you add to a meal to fill its biggest nutritional gap?

Description

An interactive **meal builder** with live nutritional analysis:

Left panel: Food Shelf

- A grid of ~80-100 curated foods, organized by food group.

- Each food is represented as a **small glyph** — a mini star/radar shape showing its nutritional fingerprint at a glance.
- Foods can be **dragged onto the plate** in the center.

Center: The Plate

- A circular plate area where dragged foods land.
- As foods are added, a **large radar chart** overlaid on the plate updates in real-time.
- The radar axes = key nutrient groups: Protein, Carbs, Fat, Fiber, Vitamins, Minerals.
- A **translucent ideal shape** (based on DRI values) shows the "target".
- **Color coding**: Green (meet/exceed DRI), amber (close), red (deficient).

Right panel: Analysis

- Live **stacked bar chart** showing each nutrient's % of daily recommended intake.
- A **"Biggest Gap" indicator** suggests which food to add next.
- A **"Hidden Interactions" alert** lights up when foods on the plate have synergy/antagonism pairs (linking back to Viz 1).

What Makes It Creative

- **Gamification**: The radar chart acts as a "score".
- **"Shape of Change" inspired glyphs**: Visual encoding of complex multi-dimensional data.
- **Real-time feedback loop**: Responds instantly to every food added/removed.
- **Bridge to Viz 1**: Direct links to interaction details from the network graph.

Data Sources

Source	What it provides
USDA FoodData Central	Nutritional profiles per food (~150 nutrients per item)
NCBI DRI Tables	Recommended daily allowances (the "target shape")
FDA V&M Chart	MineralsChart _{October2021.pdf} ##
Curated food list	Daily values + nutrient descriptions
	~80-100 common foods with pre-fetched USDA data

Chart Type: Interactive drag-and-drop builder + live radar chart + stacked bar chart + small-multiple glyphs

Visualization 3: "What the World Is Missing"

Core Question

"Where in the world are specific nutrient deficiencies most prevalent, and do the countries with the worst deficiencies actually grow foods that could solve the problem?"

Description

A **scrollytelling narrative** that guides the user through the global nutrition landscape.

Section 1: "The Hidden Hunger" (scroll intro)

- Opening statistic: *"2 billion people worldwide suffer from micronutrient deficiencies"*
- Animated counter showing affected populations.

Section 2: The Map (main visualization)

- A **choropleth world map** colored by prevalence of a selected nutrient deficiency.
- **Dropdown selector:** Iron (anaemia) — Vitamin A — Zinc — Iodine.
- *Note: Vitamin D was excluded — no standardized global dataset exists for Vitamin D deficiency by country.*

Section 3: The Paradox Panel (linked detail)

- When a country is clicked, a **side panel** opens showing:
 1. **"What this country produces"** — bar chart of top 10 locally produced crops/foods (from FAOSTAT production data, covering 245 countries).
 2. **"What these foods contain"** — nutrient breakdown of those crops (linked to USDA/Viz 2 nutrient data).
 3. **"The paradox"** — visual comparison: *"This country produces X tonnes of iron-rich crops, yet Y% of the population is anaemic."* Compares production of nutrient-rich foods against deficiency prevalence.
 4. **Trend sparkline** (deficiency rate over time).

Data Feasibility Note

Important distinction: production vs. consumption. Per-country *consumption* data (what people actually eat at the nutrient level) is scarce — very few countries run national dietary surveys. However, the core "paradox" narrative does **not require consumption data**. It works by comparing:

- **Deficiency prevalence** (well-covered: WHO, Our World in Data — 130–190 countries) against
- **Food production data** (well-covered: FAOSTAT Food Balance Sheets — 245 countries, crops + livestock).

The story becomes: *"Does this country grow foods that **could** solve its deficiency problem?"* — which is answerable and arguably a more compelling paradox than consumption-based analysis.

Data Sources

Source	What it provides
Our World in Data	Deficiency prevalence by country over time
FAOSTAT Food Balance Sheets	Food production and supply per country (245 countries, crops + livestock)
Global Nutrition Report	Country-level nutrition profiles
WHO Global Health Observatory	Anaemia, Vitamin A deficiency data
UN Data	Trade and production statistics

Summary: 3 Visualizations at a Glance

	Viz 1	Viz 2	Viz 3
Name	The Nutrient Cosmos	The Perfect Plate	What the World Is Missing
Scale	Micro (molecules)	Personal (your meal)	Macro (global)
Primary Chart	Force-directed network	Interactive builder + radar	Choropleth map + scrollytelling
Unique Angle	Chain reactions, Meal X-Ray	Gamified balance challenge	The Paradox

Technical Stack

- **Next.js:** App framework, routing, SSR
- **D3.js:** Network graph, choropleth map, custom glyphs
- **Recharts / Chart.js:** Radar charts, bar charts, sparklines
- **Framer Motion:** Page transitions, scroll animations

All Links & References

Primary Data Sources

- [USDA FoodData Central](#) — Free API, 380k+ foods with detailed nutritional profiles
- [OpenNutrition](#) — Aggregates USDA + Canadian + Danish + Australian data, 300k+ items
- [Open Food Facts](#) — Community database, 3M+ products, free JSON API
- [FooDB](#) — Deep food chemistry, 800+ foods, 70k+ compound entries
- [FAOSTAT](#) — UN food trade, production, import/export data by country
- [Our World in Data — Food](#) — Food systems, dietary patterns datasets
- [Our World in Data — Micronutrient Deficiency](#) — Deficiency prevalence by country
- [WHO Global Health Observatory](#) — Anaemia, Vitamin A deficiency data
- [Global Nutrition Report](#) — Country-level nutrition profiles
- [UN Data](#) — Global statistics on trade, agriculture, economy

Reference Tables (DRI / Daily Values)

- [DRI — Vitamins](#) — Recommended daily allowances for all vitamins by age/sex
- [DRI — Elements \(Minerals\)](#) — Recommended daily allowances for minerals
- [DRI — Macronutrients](#) — DRIs for total water and macronutrients
- [Estimated Average Requirements](#) — Estimated average requirements for nutrients
- [Acceptable Macronutrient Distribution Ranges](#) — Recommended macro ratios
- [Tolerable Upper Intake Levels — Vitamins](#) — Maximum safe vitamin intake
- [Tolerable Upper Intake Levels — Elements](#) — Maximum safe mineral intake
- [ODS Nutrient Recommendations Hub](#) — Index page for all DRI tables
- [FDA Vitamins & Minerals Chart \(PDF\)](#) — Daily values, food sources, health functions

Nutrient Interactions (Synergy / Antagonism)

- [Deanna Minich — Vitamin-Mineral Interactions Chart \(PDF\)](#) — Matrix chart of all interactions
- [Linus Pauling Institute — Micronutrient Information](#) — Evidence-based monographs per nutrient
- [NIH ODS Fact Sheets](#) — Documented interactions per nutrient

Tools / Code References

- [MCP OpenNutrition \(GitHub\)](#) — MCP server wrapping OpenNutrition, 300k+ food items
- [MCP OpenNutrition — Commit \(data structure reference\)](#)
- [SparkyFitness \(Reddit\)](#) — Self-hosted MyFitnessPal alternative, UI inspiration

Visualization Inspiration

- [Lindsey Poulter — Dashboard of the Present Future](#) — Grand Prize WDVP 2023
- [The Shape of Change](#) — 2nd Place WDVP 2023, scrollytelling + guessing game
- [Sustainability Cosmos](#) — 3rd Place WDVP 2023, dark cosmic aesthetic
- [Four Seasons](#) — Interactive Winner WDVP 2023, tree metaphor
- [WDVP 2023 Winners Overview](#) — All winners
- [IIB Awards 2019 Showcase](#) — Archive of award-winning dataviz
- [Information is Beautiful — Datasets](#) — Curated datasets collection

General Open Data Portals

- [Awesome Public Datasets \(GitHub\)](#) — Curated list of free datasets
- [EU Open Data Portal](#) — European open data
- [Data.gov](#) — US government open data
- [Socrata API](#) — API platform for government/public data
- [Wikipedia Statistical Datasets](#) — Index of statistical datasets